The C++ Programming Language

Beginner Study Notes and Practice Guide

This short guide introduces core concepts and provides practice ideas. It is an original learning summary, not a substitute for a complete textbook.

Introduction

C++ is a general-purpose programming language used for systems software, games, desktop applications, and performance-sensitive programs.

Basic program structure

A simple C++ program includes **#include <iostream>**, a **main()** function, and output such as **std::cout << "Hello";**.

Variables and types

Common types include int for integers, double for decimal values, char for a single character, bool for true/false, and std::string for text.

Conditions and loops

Use if/else to choose between paths. Use for and while loops to repeat work while a condition is true.

Functions

Functions group reusable instructions. A function can accept parameters and return a value.

Classes and objects

A class defines a type with data and functions. An object is an instance of that class.

Practice

Write a program that asks for two numbers and prints their sum. Then create a class named Book with title and price fields.

Practice checklist

- Review the concepts and write a small example.
- Run your example and test an edge case.
- Explain the solution in your own words.